09 - Business Interpretation, Results and Limitations

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. Repository's selected model

The author selects Random Forest with SMOTEENN because it has the strongest reported overall metrics among the tried configurations. The model is serialized to model.sav using pickle. The repository includes app.py, which loads the saved estimator and obtains predictions for entered customer attributes.

```
import pickle
pickle.dump(model_rf_smote, open("model.sav", "wb"))
loaded_model = pickle.load(open("model.sav", "rb"))
```

Important: The application code recreates dummy columns by concatenating the input row with first_telc.csv and applying get_dummies. This can be fragile if category levels or column ordering differ from training. A fitted preprocessing-and-model pipeline is safer. Deployment is outside the requested interview scope, but this detail helps explain the artifact.

2. Business observations

Observed characteristic	Notebook association	Possible business question
Month-to-month contract	Higher churn	Can suitable longer-term plans improve retention?
No online security / tech support	Higher churn	Are customers aware of included support/security options?
Electronic check	Higher churn	Is payment friction or customer profile a confounder?
Short tenure	Higher churn	Would onboarding support improve early retention?
Higher monthly charges	Higher churn noted	Is perceived value or price sensitivity involved?

These patterns generate hypotheses for business teams; they do not prove that changing a feature will change churn. For example, an offer or service intervention should be evaluated through a randomized or otherwise credible causal design.

3. Limitations

1. Resampling leakage: SMOTEENN is run before train-test split, compromising the post-resampling test metrics.
2. No stratification or fixed random seed in the original split.
3. No demonstrated cross-validation.
4. No systematic hyperparameter tuning; values are manually specified.
5. No documented threshold optimization or probability calibration.
6. No comprehensive baseline comparison, such as Logistic Regression.

7. No systematic final-model feature-importance or SHAP analysis.
8. Potential fragility in separately recreating one-hot columns in the application.
9. EDA observations are associative and not causal.

4. Stronger experimental plan

10. Clean and validate the original dataset; inspect the 11 missing TotalCharges records.
11. Create a stratified train/test split before any learned transformations or resampling.
12. Build a baseline with Logistic Regression and a tree model.
13. Use a preprocessing + resampling + classifier pipeline. Consider categorical-aware resampling where suitable.
14. Compare Decision Tree, Random Forest, and a boosting model through stratified cross-validation.
15. Tune hyperparameters using a metric aligned with business costs.
16. Choose a threshold using validation predictions and an explicit cost/benefit framework.
17. Evaluate once on the untouched test set; report precision, recall, F1, PR-AUC, ROC-AUC, confusion matrix, and calibration as appropriate.

5. Interview-ready explanation

The repository selects Random Forest with SMOTEENN based on its reported metrics. The EDA suggests that contract type, tenure, payment method, and service subscriptions are associated with churn. The key limitation is that resampling was performed before splitting, so the high resampled test metrics are not trustworthy. I would correct that with a stratified holdout, a leakage-safe pipeline, cross-validation, systematic tuning, and threshold selection based on the cost of missed churners versus unnecessary retention outreach.

6. Questions to prepare

Q. Can an important feature be a cause of churn?

Not from predictive importance or correlation alone. Importance measures model reliance or association, not causal effect.

Q. What is permutation importance?

It measures the drop in a model metric after randomly permuting a feature, breaking its relationship with the target while leaving other columns unchanged.

Q. What is SHAP?

A framework that attributes a prediction to feature contributions using Shapley-value concepts. It explains model output, not causal effects.

Q. Why choose a threshold other than 0.5?

The best threshold depends on class prevalence, probability calibration, and relative costs of false positives and false negatives.

Q. What would you say if an interviewer challenges the reported 94% accuracy?

I would clarify it is the repository's reported score, but because SMOTEENN was applied before the split, it is contaminated by leakage. I would not claim it as unbiased test performance and would rerun the experiment correctly.